Premise Design Prior to Linguistic Projection and Premise Updating as Projection Switching

An Integrated Theory Based on Semantic Fields, Phase-Preserving Representations, Reinjection Possibility, and Search-Space Control

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Abstract

This paper aims to unify, within a consistent symbolic framework, the premise extraction layer, phase-preserving representation, linguistic projection, reinjection possibility, and premise updating, while maintaining the view that premises are not sets of natural-language sentences but constraint structures embedded in a higher-order semantic field.

Existing theories of premise design have argued that premises exist not as propositional sequences but as sets of constraints containing contextual bindings, priorities, exception conditions, responsibility histories, and suspended components, and that natural language is not their substance but merely a purpose-dependent linear projection of them. However, such discussions have primarily focused on the definition of static structures, and have not sufficiently formalized by what dynamic mechanism premise updating occurs, nor how linguistic projection functions not merely as a final output but as part of search-space control.

This paper first introduces the structural space $\mathcal{S}$ as the mother space of higher-order semantic structures, and defines the semantic field $\Psi_i(t)$ as the local state occupied by subject i at time t. On this basis, it redefines the premise extraction operator E_i, the phase-preserving representation operator M, the purpose-dependent linguistic projection P_ω, the reinjection operator Q, the semantic density ρ, and the semantic-density preservation rate $\eta_i^{(\omega)}$. Furthermore, in order to resolve the previous ambiguity surrounding the symbol Φ, this paper clearly separates the phase-preserving representation as $\Xi_i(t)$, the projection-switching operator as Φ_Π, and the in-projection structural update operator as $\Phi_\mathcal{S}$.

The central claim of this paper is that premise updating is neither sentence revision nor a simple state transition, but a composite process consisting of (i) collapse detection on phase-preserving representations, (ii) switching perspectives

over a set of projections, and (iii) structural updating under a new projection. This makes it possible to preserve the previous claim that "language is not the substance but a projection," while simultaneously introducing the dynamic understanding that "linguistic projection is also an operation that carves out the search space."

Ultimately, this paper presents an implementable theoretical framework for AI, education, institutional design, and high-density thought operation by integrating the premise extraction layer as a static preservation structure and projection switching as a dynamic operational mechanism.

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1. Introduction

1.1. Problem Setting

In advanced thought, dialogue, creativity, institutional design, and AI operation, what is essential is not only "what was said," but also "under what premise structure it is being said." Yet in many cases, premises are subsumed under vague terms such as beliefs, values, positions, and ideologies, and have not been sufficiently treated as implementable structures.

As a result, premise updating is often understood as a paraphrase of sentences or a change in explanation. In reality, however, prior to such linguistic change, at least the following are changing: (i) which constraints are effective, (ii) which relations are preserved, and (iii) under what phase the object is being observed. Therefore, to genuinely address premise updating, a theory is needed that treats semantic fields, constraints, phase-preserving representations, and projection switching in an integrated manner, rather than merely sentence-level differences.

1.2. Purpose of This Paper

The purpose of this paper is the following fivefold:

1. To integrate the existing theory of the premise extraction layer and the theory of premise updating as projection switching within a non-contradictory symbolic system.

2. To preserve the static claim that "language is not the substance but a projection," while formalizing the dynamic claim that "projection is also search-space control."

3. To define the premise extraction layer as a structure that obtains a constraint set $C_i(t)$ and a phase-preserving representation $\Xi_i(t)$ from the semantic field $\Psi_i(t)$.

4. To define premise updating as a composite of the projection-switching operator Φ_Π and the in-projection structural update operator $\Phi_\mathcal{S}$.

5. To present a search-space control model that can be connected to implementation in AI.

1.3. Basic Position of This Paper

The basic position of this paper may be summarized in the following sentence:

> Premises do not exist as natural-language sentences, but as constraint structures embedded in semantic fields over a structural space; natural language is a purpose-dependent projection of such structures, and premise updating is a composite process of constraint reorganization and projection switching.

2. The Basic Problem: Why Language Alone Is Not Enough

2.1. Premises Are Not Sentences

Behind any judgment there operate simultaneously not a single proposition but at least the following elements:

- scope of application
- priority ordering
- exception conditions
- evidential history
- suspended components
- contextual dependency
- update conditions

Accordingly, even if a premise can be expressed in a sentence, it does not exist as a sentence. A premise is rather a set of constraints that governs selection, connection, rejection, and suspension.

2.2. Natural Language Is a Linear Projection

Natural language is a one-dimensional sequence with temporal order, and outputs structures that originally existed simultaneously and in parallel by arranging them into an order. As a result, at least the following distortions arise:

(a) compression distortion: multilayered constraints are compressed into a short symbol sequence;

(b) order distortion: simultaneously existing elements acquire before/after relations;

(c) granularity distortion: fundamental principles and local rules appear on the same level;

(d) context stripping: application conditions and suspended components drop out.

Thus, natural language is not only a medium for transmitting meaning but also a medium that can collapse meaning.

2.3. Summarization and Extraction Are Different

Summarization is an operation that reduces the amount of expression. By contrast, extraction is an operation that identifies constraint structures that are effective for operation. What premise design requires is not a summarizer but a constraint extractor. If this point is overlooked, premise updating is mistakenly reduced to sentence rewriting.

3. Unification of the Symbolic System

In order to connect the existing definitions and the new dynamic model consistently, this paper adopts the following notation:

- $\mathcal{S}$: the mother space of higher-order semantic structures (structural space)
- $\Psi_i(t)$: the semantic field of subject i at time t
- $R_i(t)$: the update history / responsibility history of subject i
- $C_i(t)$: the set of premise constraints of subject i
- E_i: the premise extraction operator
- $\Xi_i(t)$: the phase-preserving representation
- M: the operator that constructs a phase-preserving representation from a constraint set
- Π: the set of available projections
- $\pi \in \Pi$: an individual projection
- P_ω: the linguistic projection corresponding to purpose ω
- $L_i^{(\omega)}(t)$: the linguistic output for purpose ω
- Q: the reinjection (re-parsing) operator from language to the phase-preserving side
- $\rho(\cdot)$: semantic density
- $\eta_i^{(\omega)}$: semantic-density preservation rate
- Φ_Π: the projection-switching operator
- $\Phi_{\mathcal{S}}^{(\pi)}$: the structural update operator under projection π

Remark 3.1. *In earlier drafts, Φ referred both to the "phase-preserving representation" and to the "update operator," causing a symbolic collision. In this paper, the phase-preserving representation is separated as $\Xi_i(t)$, and the update operator as Φ_Π and $\Phi_{\mathcal{S}}$, thereby resolving the conflation between the static layer and the dynamic layer.*

4. Structural Space and Semantic Field

Definition 4.1 (Structural Space)**.** *The structural space $\mathcal{S}$ is the mother space of higher-order semantic structures, including concept bundles, relation bundles, narrative bundles, role bundles, contextual bindings, responsibility histories, exception conditions, suspended components, and the connectivity structures among them.*

Definition 4.2 (Semantic Field)**.** *The semantic field $\Psi_i(t)$ of subject i at time t is the local semantic configuration state occupied by that subject on the structural space $\mathcal{S}$, and is an updatable structural field containing judgments, utterances, silences, suspensions, rejections, expectations, responsibility histories, and unlinguized constraints.*

It may be written as

$$\Psi_i(t) \in \mathcal{S}$$

or, when the semantic field is interpreted as a local region,

$$\Psi_i(t) \subset \mathcal{S}.$$

This paper adopts the former as basic while permitting the latter as a local-regional interpretation.

Remark 4.3. *$\mathcal{S}$ is the upper-level mother space of the overall theory, whereas $\Psi_i(t)$ is a subject-dependent and time-dependent concrete state. This distinction makes it possible to separate "structure itself" from "the semantic configuration currently active at that moment."*

5. Definition of the Premise Extraction Layer

Definition 5.1 (Set of Premise Constraints)**.** *The set of premise constraints $C_i(t)$ for subject i at time t is the set of constraints extracted from the semantic field $\Psi_i(t)$ and the history $R_i(t)$ that actually operate in judgment, connection, rejection, and suspension, and is expressed as*

$$C_i(t) = \{c_{i1}(t), c_{i2}(t), \ldots, c_{in}(t)\}.$$

Each constraint $c_{ik}(t)$ has at least a scope, a priority, exception conditions, update conditions, an evidential history, and suspended components.

Definition 5.2 (Premise Extraction Operator)**.** *The premise extraction operator E_i is the action that obtains the set of premise constraints $C_i(t)$ from the semantic field $\Psi_i(t)$ and the history $R_i(t)$, and is defined as*

$$E_i : (\Psi_i(t), R_i(t)) \to C_i(t).$$

Definition 5.3 (Phase-Preserving Representation)**.** *The phase-preserving representation $\Xi_i(t)$ is an intermediate representation that preserves, without collapse, the dependency*

relations, hierarchy, conflict relations, exceptions, scopes of application, and suspended components of the constraint set $C_i(t)$, and is defined as

$$\Xi_i(t) = M(C_i(t)).$$

Here M permits any phase-preserving form of representation, including graphs, bundles, tensors, directed constraint networks, and hierarchical bundles.

Thus, the static premise extraction layer has the following flow:

$$\Psi_i(t), R_i(t) \xrightarrow{E_i} C_i(t) \xrightarrow{M} \Xi_i(t).$$

6. Linguistic Projection and Reinjection Possibility

Definition 6.1 (Purpose-Dependent Linguistic Projection)**.** *Let P_ω denote the linguistic projection for purpose ω, and define the linguistic output $L_i^{(\omega)}(t)$ of subject i at time t by*

$$L_i^{(\omega)}(t) = P_\omega(\Xi_i(t)).$$

Definition 6.2 (Reinjection Possibility)**.** *A linguistic output $L_i^{(\omega)}(t)$ is said to be reinjectable if, via the reinjection operator Q,*

$$Q(L_i^{(\omega)}(t)) \approx \Xi_i(t)$$

holds to a sufficient degree. Here $\approx$ does not mean exact identity, but requires at least that core constraints, priority relations, exception conditions, and suspended components have not been fatally lost.

Definition 6.3 (Semantic Density)**.** *Semantic density ρ is an abstract quantity expressing the density of semantic binding force, dependency relations, operational constraints, and reinjectable structure retained by a representation, and is regarded as*

$$\rho : \mathcal{S} \cup \mathrm{Im}(M) \cup \mathrm{Im}(P_\omega) \to \mathbb{R}_{\geq 0}.$$

Definition 6.4 (Semantic-Density Preservation Rate)**.** *The semantic-density preservation rate $\eta_i^{(\omega)}$ for purpose ω is defined by*

$$\eta_i^{(\omega)} = \frac{\rho\big(Q(L_i^{(\omega)}(t))\big)}{\rho\big(\Xi_i(t)\big)}.$$

Ideally, $\eta_i^{(\omega)} \to 1$ is desirable, but in practice the acceptable loss differs by purpose.

7. Redefining Language as Projection

The symbols introduced up to the previous section follow the existing theory of the premise extraction layer. We now take a further step and regard language not as the "final output" but as a projection that makes part of the structure visible and operable.

Definition 7.1 (Set of Projections). *Let Π be the set of available projections. Each $\pi \in \Pi$ is a mode of observation and operation over the structural space, and may include, for example, descriptive projection, role projection, narrative projection, institutional projection, and identity projection.*

Definition 7.2 (General Projection). *A general projection π is a mapping that yields the visible and operable section of the structural space or phase-preserving representation, and is written as*

$$\pi : \mathcal{S} \text{ or } \mathrm{Im}(M) \to \mathcal{O}_\pi.$$

The purpose-dependent linguistic projection P_ω is a special case of this general projection.

Remark 7.3. *The important point here is that while preserving the earlier claim that language is not the substance but a projection, we add the dynamic meaning that a projection is not merely a display, but determines the visible region and the space of possible search.*

Definition 7.4 (Reachable Region). *The structural region searchable under projection π is defined as*

$$\mathcal{A}_\pi \subseteq \mathcal{S}.$$

This is the subset of structures that are visible, usable, and operable under that projection.

8. Axiomatic Conditions

Axiom 8.1 (Premises Are Constraints, Not Propositions). *Premises exist not as single propositions subject to truth evaluation, but as sets of constraints governing connection, selection, rejection, and suspension.*

Axiom 8.2 (Natural Language Is Not the Substance but a Projection). *Natural-language sentences are not the substance of premises, but only purpose-dependent projections of phase-preserving representations.*

Axiom 8.3 (Silence and Suspension Are Not Deficiencies). *Unlinguized elements, silence, suspension, and undefinedness are not deficiencies but constituent elements of the semantic field and the constraint structure.*

Axiom 8.4 (Extraction Takes Priority over Summarization). *In premise operation, extraction of constraint structure takes priority over simplification of sentences.*

Axiom 8.5 (Premise Updating Is Constraint Reorganization, Not Sentence Revision). *Premise updating occurs not as sentence rewriting, but as reorganization of the constraint set and the phase-preserving representation.*

Axiom 8.6 (Premise Updating Includes Projection Switching)**.** *Premise updating includes, in addition to constraint reorganization, switching the projection under which a structure is observed and operated.*

9. Premise Updating as Projection Switching

We now define the dynamic core of this paper.

Definition 9.1 (Projection-Switching Operator)**.** *The projection-switching operator* Φ_Π *is the operator that selects the next projection* π_{t+1} *on the basis of the current phase-preserving representation* $\Xi_i(t)$ *and the current projection* π_t*, and is defined as*

$$\Phi_\Pi : (\Xi_i(t), \pi_t) \mapsto \pi_{t+1}.$$

Definition 9.2 (In-Projection Structural Update Operator)**.** *The structural update operator* $\Phi_{\mathcal{S}}^{(\pi)}$ *under projection* π *is the action that updates the current structural state or phase-preserving representation within that projection, and is defined as*

$$\Phi_{\mathcal{S}}^{(\pi)} : \Xi_i(t) \mapsto \Xi_i(t+1)$$

or, when necessary,

$$\Phi_{\mathcal{S}}^{(\pi)} : \Psi_i(t) \mapsto \Psi_i(t+1).$$

Definition 9.3 (Integrated Form of Premise Updating)**.** *Premise updating is defined as the composite of projection switching and in-projection updating:*

$$\begin{cases} \pi_{t+1} = \Phi_\Pi(\Xi_i(t), \pi_t), \\ \Xi_i(t+1) = \Phi_{\mathcal{S}}^{(\pi_{t+1})}(\Xi_i(t)). \end{cases}$$

This is not a mere state revision, but a composite process in which one:

1. observes a structure under a given projection,
2. detects collapse or saturation within that projection,
3. switches to a more appropriate projection, and
4. again performs structural updating under the new projection.

10. Redefining Convergence

The conventional notion of convergence is redefined here in two layers.

Definition 10.1 (In-Projection Convergence)**.** *Under a projection* π*, if*

$$\Xi^* = \Phi_{\mathcal{S}}^{(\pi)}(\Xi^*)$$

holds, then Ξ^* is called an in-projection fixed point under projection π.

Definition 10.2 (Projection Convergence)**.** *For a state Ξ^* and a projection π^*, if*

$$\pi^* = \Phi_\Pi(\Xi^*, \pi^*)$$

holds, then π^* is called a projection fixed point for that state.

Definition 10.3 (Complete Convergence)**.** *A pair (Ξ^*, π^*) is said to exhibit complete convergence if it simultaneously satisfies*

$$\begin{cases} \Xi^* = \Phi_{\mathcal{S}}^{(\pi^*)}(\Xi^*), \\ \pi^* = \Phi_\Pi(\Xi^*, \pi^*). \end{cases}$$

Remark 10.4. *By this definition, convergence is no longer simply the "stabilization of a point," but includes the question of under which perspective no further switching is necessary.*

11. Collapse and Deletion Rules

Projection switching becomes necessary when, under the current projection, the structure collapses or semantic-density improvement reaches saturation.

Definition 11.1 (Invariants)**.** *Let*

$$I_1, I_2, \ldots, I_m$$

be the structural invariants that must be preserved in premise operation. Representative examples include:

- *consistency invariant*
- *historical continuity invariant*
- *impact-effectiveness invariant*
- *non-redundancy invariant*
- *suspension-preservation invariant*

Definition 11.2 (Collapse)**.** *A state Ξ is said to be in collapse if*

$$\exists j \ s.t. \ I_j(\Xi) = 0$$

holds.

Definition 11.3 (Deletion Rule)**.** *A deletion rule is a rule that removes from the operational space those candidates for which an invariant is broken, and defines the valid region as*

$$\mathcal{C}_{valid} = \{\Xi \mid I_j(\Xi) = 1 \text{ for all relevant } j\}.$$

Proposition 11.4. *Collapse detection is a principal trigger for projection switching.*

Proof. If the candidate set obtained under a projection π_t continuously violates invariants, then the reachable region $\mathcal{A}_{\pi_t}$ of that projection either fails to contain sufficient valid solutions or the way of observing is itself inappropriate. Hence, switching to another projection π_{t+1} that yields greater semantic density or a lower collapse rate becomes rational. □

12. Propositions and Corollaries

Proposition 12.1. *A design that directly performs linguistic projection P_ω from the semantic field $\Psi_i(t)$ without passing through the phase-preserving representation $\Xi_i(t)$ is likely to damage the dependency structure of premises.*

Proof. $\Psi_i(t)$ is a higher-dimensional parallel structure, whereas P_ω is a projection into a one-dimensional sequence or other low-dimensional image. Without passing through an intermediate phase-preserving representation, one cannot localize which dependency relations were lost, and structural degeneration becomes invisible. □

Proposition 12.2. *A linguistic expression lacking reinjection possibility may function as an explanation, but is insufficient as an internal operational language for a premise-updating OS.*

Proof. Internal operation requires that a linguistic expression be returned to constraints, priorities, exception conditions, and suspended components. An expression with low reinjection possibility may serve for external explanation, but cannot be used for update logs or difference management. □

Proposition 12.3. *Different linguistic projections P_ω are necessary for different purposes.*

Proof. Even given the same phase-preserving representation $\Xi_i(t)$, the structural elements that must be preserved differ across internal operation, external explanation, education, signature, institutional connection, and so on. Therefore a single natural-language output cannot satisfy all purposes. □

Corollary 12.4. *The relation between an original text and a translation should not be regarded as that between one authentic original and its degraded copy, but rather as different reprojections from a common phase-preserving representation.*

Proposition 12.5. *Premise updating consists not only in reorganization of the constraint set and the phase-preserving representation, but also includes switching perspectives over the set of projections.*

Proof. Sentence differences alone cannot describe update impossibility, the reason for collapse, or the cause of search saturation. By contrast, if one tracks the constraint set $C_i(t)$ and the phase-preserving representation $\Xi_i(t)$, and switches the projection π_t when necessary, updating can be managed as structural difference. □

13. Implementation Model: Premise Updating as Search-Space Control

To connect this theory to implementation, it is not necessary to maintain the whole structural space $\mathcal{S}$ in full. In practice, it is sufficient to approximately maintain a phase-preserving representation Ξ and a set of projections Π.

13.1. Minimal Internal Representation

A state may be approximately represented by a data structure such as:

```
State Xi:
  constraints
  relations
  priorities
  exceptions
  history
  suspensions
```

A projection may be represented as:

```
Projection pi:
  visible_constraints
  visible_relations
  weighting
  forbidden_patterns
  projection_goal
```

13.2. Minimal Algorithm

```
Xi = initial_phase_preserving_representation
pi = initial_projection

while True:
    candidates = generate_candidates(Xi, pi)
```

```
candidates = filter_by_invariants(candidates, pi)
scored = score_by_meaning_density(candidates, pi)

best = select_best(scored)

if not improves(best, Xi, pi):
    new_pi = switch_projection(Xi, pi)

    if new_pi == pi:
        break

    pi = new_pi
    continue

Xi = best
```

13.3. Mathematical Approximation

If this procedure is approximated mathematically, then

$$\pi_{t+1} = \Phi_{\Pi}(\Xi_t, \pi_t), \qquad \Xi_{t+1} = \Phi_{\mathcal{S}}^{(\pi_{t+1})}(\Xi_t)$$

where these respectively represent:

- **switching**: which perspective to move to;
- **updating**: how to reorganize within that perspective.

14. Implications for AI, Education, and Institutional Design

14.1. Implications for AI

Many current AI systems depend heavily on correspondences between input and output sentences and on token-sequence prediction accuracy. From the standpoint of this paper, however, that alone is insufficient for premise-updating capability. What is required is:

- approximate estimation of the semantic field $\Psi_i(t)$,
- extraction of the set of constraints $C_i(t)$,
- maintenance of the phase-preserving representation $\Xi_i(t)$,
- operation of multiple projections Π,
- collapse detection and projection switching Φ_{Π},
- reinjectable linguistic projections P_ω.

From this perspective, premise-updating capability is redescribed not as a mystery of a higher intellect, but as a design problem of search-space control and projection switching.

14.2. Implications for Education

Many failures in education arise from compressing a learner's semantic field and premise structure into a correct-answer sentence too early. Questions, suspensions, silences, and undefinedness are not deficiencies but margins for premise formation. Therefore, what education requires is not early linguistic fixation of answers, but designs that preserve the premise extraction layer and the phase-preserving representation.

14.3. Implications for Institutional Design

Institutions fix complex premise structures into laws, rules, and commands. However, externalized language tends to possess irreversibility. Therefore, in institutional design, the phase-preserving representation Ξ prior to externalization must be sufficiently preserved, and institutional language should be designed as a purpose-dependent projection from it.

15. Discussion: The Relation Between the Static Layer and the Dynamic Layer

The integrated theory presented in this paper consists of two layers.

15.1. The Static Layer

The static layer is

$$\Psi_i(t), R_i(t) \xrightarrow{E_i} C_i(t) \xrightarrow{M} \Xi_i(t) \xrightarrow{P_\omega} L_i^{(\omega)}(t),$$

which is precisely the definition of the premise extraction layer. It shows that the substance of premises is not sentences but constraint structures, and that language is the final projection.

15.2. The Dynamic Layer

The dynamic layer is

$$\begin{cases} \pi_{t+1} = \Phi_\Pi(\Xi_i(t), \pi_t), \\ \Xi_i(t+1) = \Phi_{\mathcal{S}}^{(\pi_{t+1})}(\Xi_i(t)), \end{cases}$$

which is the operational mechanism of premise updating and convergence. Here, the very perspective under which the structure is viewed becomes an object of updating.

15.3. Integrated Understanding of the Two

Accordingly, in this paper:

- the premise extraction layer is the static OS structure that defines "what is to be preserved," while
- projection switching is the dynamic process management that defines "how to operate and when to change perspective."

In this sense, the former corresponds to memory structure, and the latter to process management.

16. Future Tasks

This paper is a definitional paper of an integrated theory, and leaves the following as future tasks:

1. a concrete measurement method for semantic density ρ,
2. a formal specification of the phase-preserving representation Ξ,
3. a classification system for the projection set Π,
4. the minimal set of collapse invariants I_j,
5. methods for designing or learning the projection-switching operator Φ_Π,
6. synchronization conditions for semantic fields across multiple subjects,
7. implementation methods for constraint-difference logs and projection-difference logs,
8. pre-externalization compression-loss inspection methods for institutional design.

17. Conclusion

This paper has maintained the position that premises are defined not as natural-language sentences but as constraint structures embedded in semantic fields, and has integrated their static structure and dynamic updating mechanism into a single symbolic system.

First, it introduced $\mathcal{S}$ as the mother space of higher-order semantic structures, and positioned $\Psi_i(t)$ as a subject-dependent local state. Second, it uniformly defined the premise extraction operator E_i, the set of constraints $C_i(t)$, the phase-preserving representation $\Xi_i(t)$, the purpose-dependent linguistic projection P_ω, the reinjection operator Q, and the semantic-density preservation rate $\eta_i^{(\omega)}$. Third, it redefined premise updating, which had previously remained ambiguous, as the composite of the projection-switching operator Φ_Π and the in-projection update operator $\Phi_\mathcal{S}$. Fourth, it reformulated convergence in two layers—"in-projection convergence" and "projection convergence"—and defined complete convergence as the simultaneous fixed point of both. Fifth, it showed that this theory provides a design principle of implementable search-space control for AI, education, and institutional design.

The final conclusion of this paper is as follows:

> To design beyond language does not mean abandoning language. It means relocating the substance of premises from natural-language sentences to semantic fields, constraint structures, and phase-preserving representations, and further understanding premise updating as the composite of projection switching and structural updating, thereby repositioning language as a purpose-dependent projection onto higher-order structure.

In this sense, the core of premise updating and high-density thought operation lies, prior to "what to say and how to say it," in "what to preserve at which layer, under which projection to operate it, and when to switch perspectives."

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